



PRODUCT NAME

Premium Leg Pain Formula

Description

A high viscosity cream designed for quick relief from leg pain with anti-inflammatory properties and fast absorption. Suitable for frequent use, providing both care and comfort for sore muscles and joints.

Technical Specifications

pH Level:	7
Viscosity:	High
Shelf Life:	24 months
Batch Size:	50-100 liters
Processing Time:	1.5 hours
Temperature:	Temperature range between 30°C to 70°C during processing
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	High-shear mixer, heating/cooling system, pH meter
Certification:	GMP (Good Manufacturing Practice)

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent and base for the cream
2	Sunflower Seed Oil	Helianthus Annuus Seed Oil	15.0%	Emollient, provides moisturizing properties
3	Glycerin	Glycerin	10.0%	Humectant, retains moisture
4	Menthol	Menthol	3.0%	Counterirritant for pain relief and cooling effect
5	Cetearyl Alcohol	Cetearyl Alcohol	4.0%	Thickener and emulsifier
6	Polysorbate 60	Polysorbate 60	2.0%	Emulsifier to stabilize ingredients
7	Methyl Salicylate	Methyl Salicylate	2.0%	Pain relief and anti-inflammatory properties
8	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative to prevent microbial growth
9	Tocopheryl Acetate	Tocopheryl Acetate	1.0%	Antioxidant and skin-conditioning agent
10	Sodium Hydroxide	Sodium Hydroxide	2.0%	pH adjuster
Total			100%	

Manufacturing Process

Phase 1: Phase A

- Combine Water and Glycerin in a main mixing tank.
- Heat to 70°C while stirring until uniform.

Phase 2: Phase B

- In a separate container, melt Sunflower Seed Oil and Cetearyl Alcohol together with Polysorbate 60 at 70°C.

2. Add the melted mixture to Phase A while maintaining the temperature and mix until emulsified.

Phase 3: Phase C

1. Cool the mixture to 40°C.
2. Add Menthol and Methyl Salicylate and mix until evenly distributed.

Phase 4: Phase D

1. At 35°C, add Phenoxyethanol and Tocopheryl Acetate.
2. Adjust pH to 7 using Sodium Hydroxide while stirring continuously.

Required Equipment

High-shear mixer, heating/cooling system, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: GMP (Good Manufacturing Practice)

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50-100 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a generous amount to the affected area and massage gently until fully absorbed. Use 2-3 times daily or as needed.